

Report Exercise: Land-Cover Classification



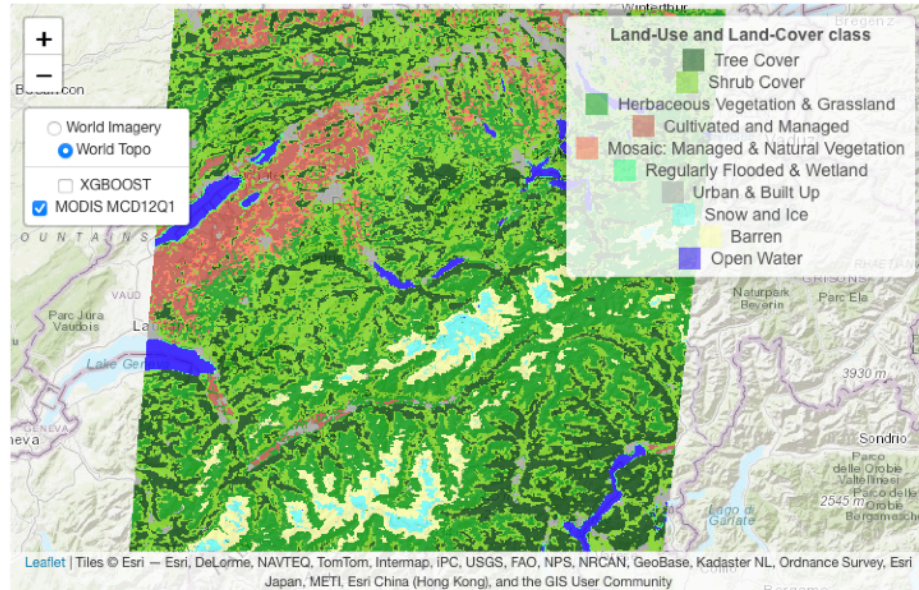
Applied Geodata Science II

Session 9

Fabian Bernhard
11.11.2025



Land-Use and Land-Cover (LULC) Classification



MODIS satellites

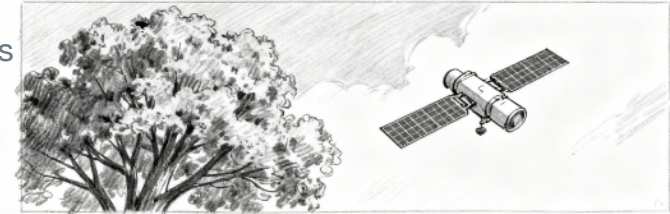
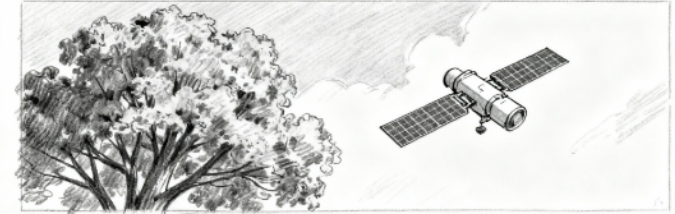
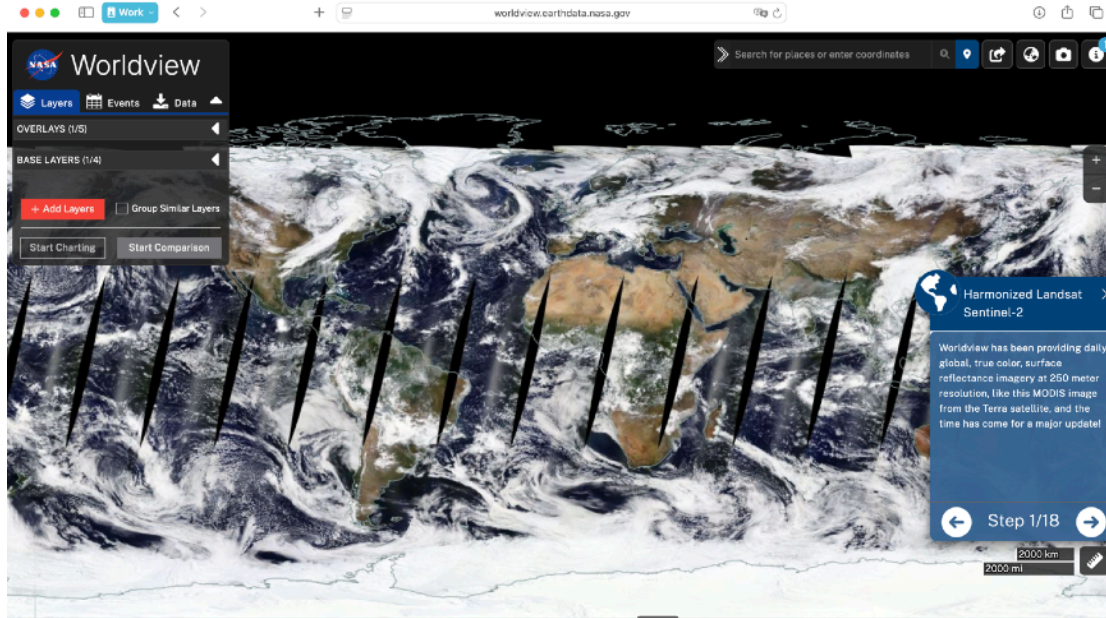


Table 1. IGBP land cover classification system.

Class	Class name	Description
1	Evergreen needleleaf forests	Lands dominated by needleleaf woody vegetation with a percent cover >60% and height exceeding 2 m. Almost all trees remain green all year. Canopy is never without green foliage.
2	Evergreen broadleaf forests	Lands dominated by broadleaf woody vegetation with a percent cover >60% and height exceeding 2 m. Almost all trees and shrubs remain green year round. Canopy is never without green foliage.
3	Deciduous needleleaf forests	Lands dominated by woody vegetation with a percent cover >60% and height exceeding 2 m. Consists of seasonal needleleaf tree communities with an annual cycle of leaf-on and leaf-off periods.
4	Deciduous broadleaf forests	Lands dominated by woody vegetation with a percent cover >60% and height exceeding 2 m. Consists of broadleaf tree communities with an annual cycle of leaf-on and leaf-off periods.
5	Mixed forests	Lands dominated by trees with a percent cover >60% and height exceeding 2 m. Consists of tree communities with interspersed mixtures or mosaics of the other four forest types. None of the forest types exceeds 60% of landscape.
11	Permanent wetlands	Lands with a permanent mixture of water and herbaceous or woody vegetation. The vegetation can be present either in salt, brackish, or fresh water.
12	Croplands	Lands covered with temporary crops followed by harvest and a bare soil period (e.g., single and multiple cropping systems). Note that perennial woody crops will be classified as the appropriate forest or shrub land cover type.
13	Urban and built-up lands	Land covered by buildings and other man-made structures.
14	Cropland/natural vegetation mosaics	Lands with a mosaic of croplands, forests, shrubland, and grasslands in which no one component comprises more than 60% of the landscape.
15	Snow and ice	Lands under snow/ice cover throughout the year.
16	Barren	Lands with exposed soil, sand, rocks, or snow and never have more than 10% vegetated cover during any time of the year.
17	Water bodies	Oceans, seas, lakes, reservoirs, and rivers. Can be either fresh or salt-water bodies.

Figure 7.5: MODIS MCD12Q1 LULC classes over an area of Switzerland (classes are simplified to those from Fritz et al. 2017)

Land-Use and Land-Cover (LULC) Classification

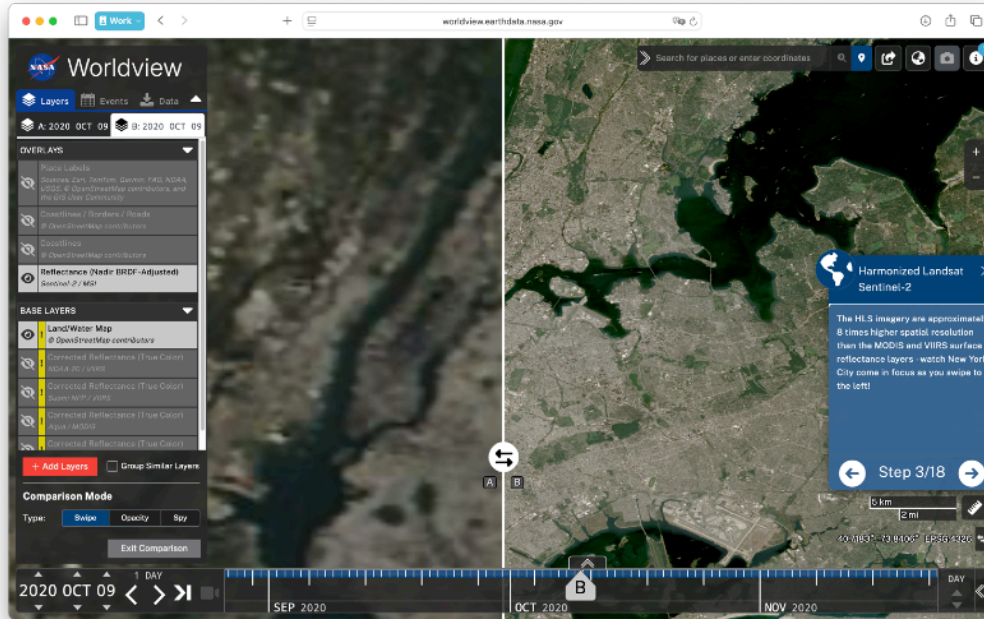
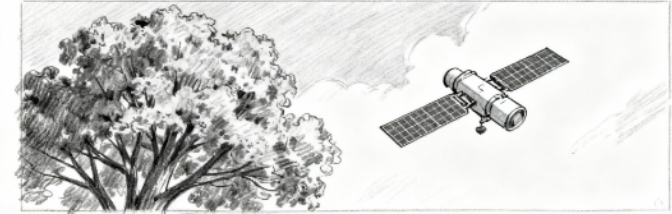


<https://worldview.earthdata.nasa.gov/>

Land-Use and Land-Cover (LULC) Classification

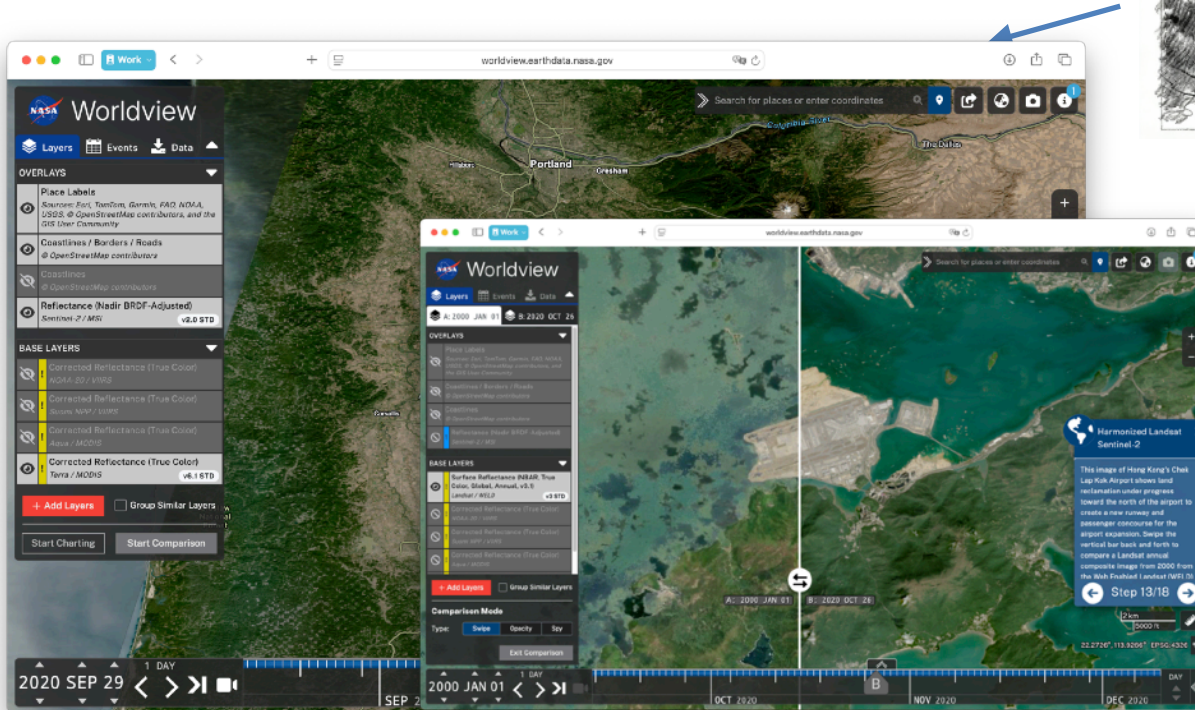
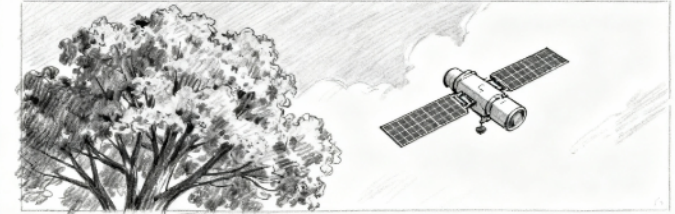
MODIS Terra
launched 1999

Sentinel-2
launched 2015



<https://worldview.earthdata.nasa.gov/>

Land-Use and Land-Cover (LULC) Classification

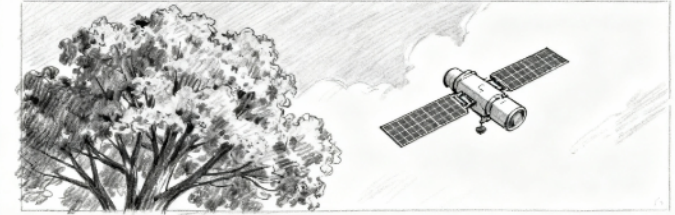
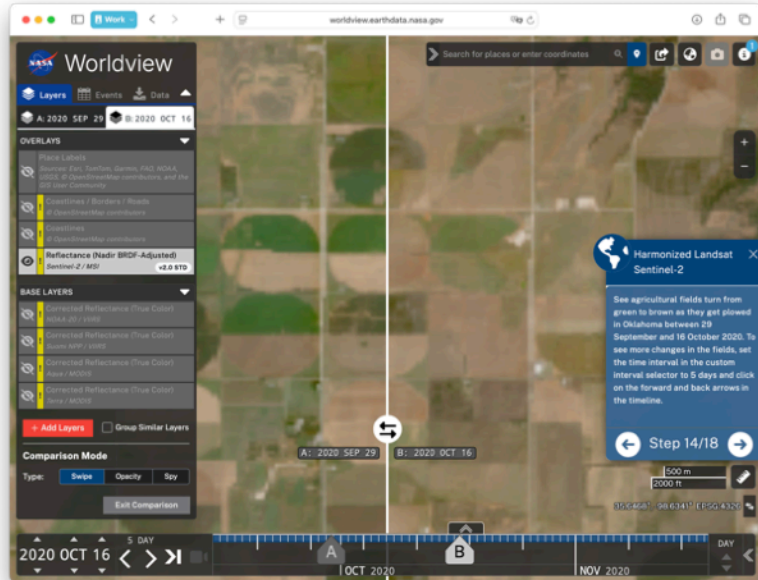


<https://worldview.earthdata.nasa.gov/>

Land-Use and Land-Cover (LULC) Classification

Before ploughing
Sep 29th

After ploughing
Oct 16th

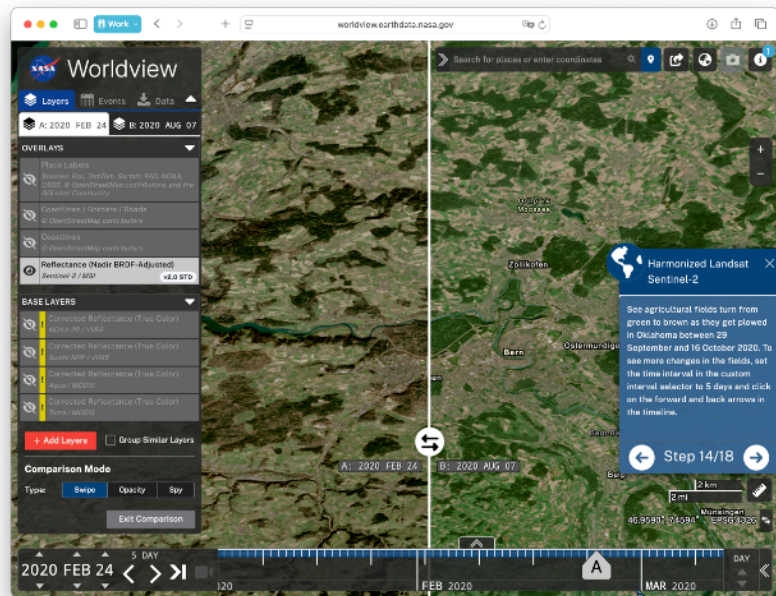
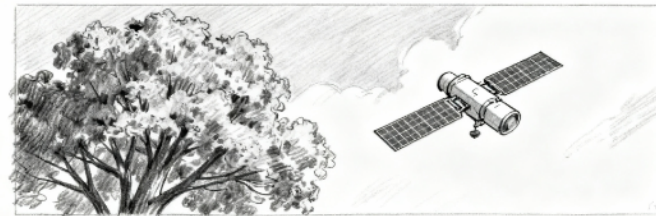


<https://worldview.earthdata.nasa.gov/>

Land-Use and Land-Cover (LULC) Classification

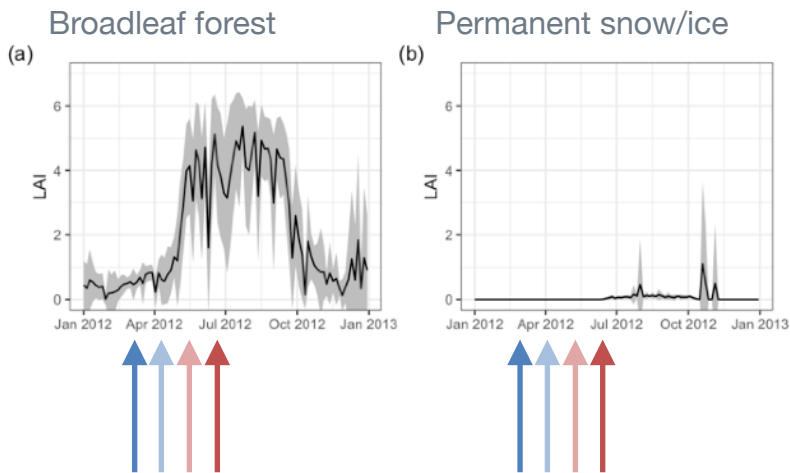
Winter
Feb 24th

Summer
Aug 07th



<https://worldview.earthdata.nasa.gov/>

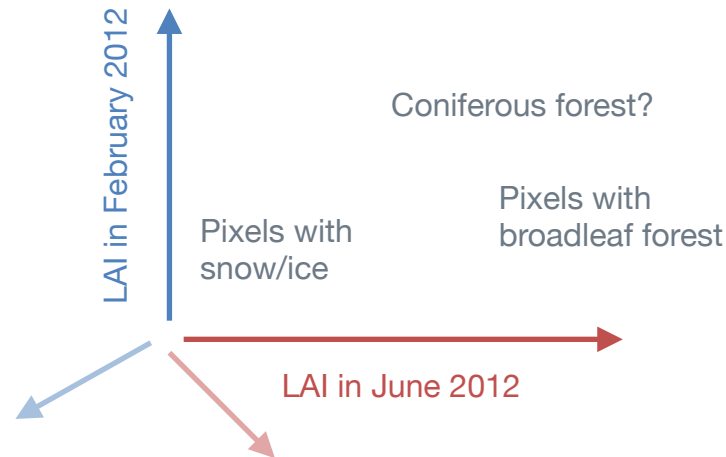
Use seasonal variation to discover land-cover type



Pixel	LAI Feb	LAI June	LAI Apr	LAI May
1	...	...	...	...
2	...	...	...	...
...	...	...	...	...
...	...	...	...	...

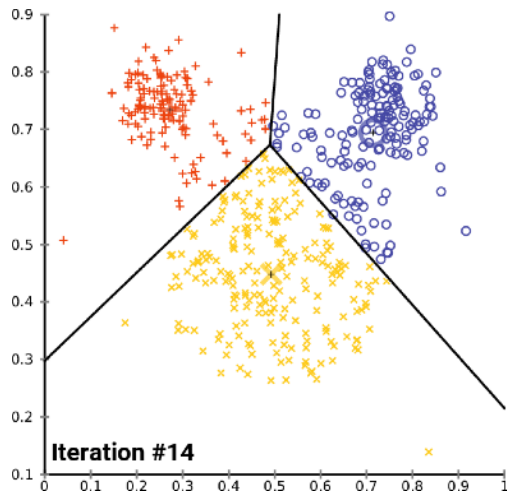
Coniferous forest

...



k-means clustering (chapter 7.1)

Single hyperparameter:
 k = number of desired classes

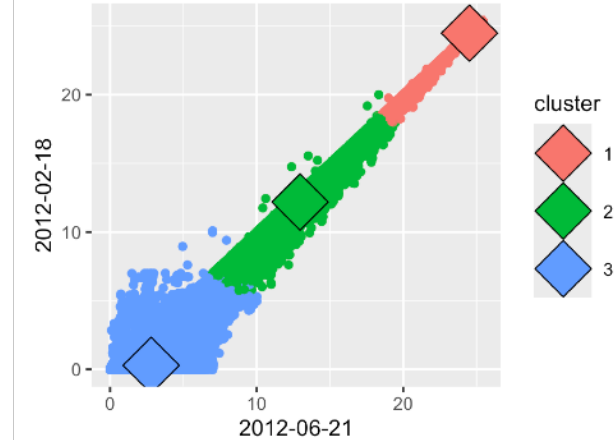
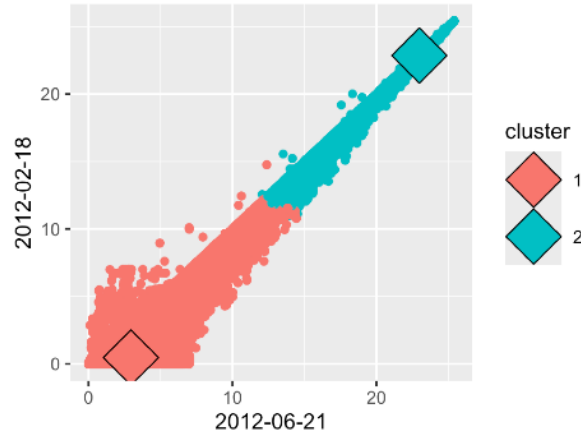


Pixel	LAI Feb	LAI Apr	LAI May	LAI June
1	...	...	...	...
2	...	...	...	...
...	...	...	...	...
...	...	...	...	...



k-means clustering (chapter 7.1)

Single hyperparameter:
 k = number of desired classes



k-means clustering compared to Boosted Trees

k-means: unsupervised classification (chapter 7.1)

Pixel	LAI Feb	LAI Apr	LAI May	LAI June
1	...	...	...	...
2	...	...	...	...
...	...	...	...	...
190'000				

Boosted classification trees: supervised classification (chapter 7.2)

Pixel	Land Cover	B1 Feb	B1 Apr	B1 May	B1 June	...	B4 Feb	B4 Apr	B4 May	B4 June	...
1	Trees	...	...	...	...	...	...	...	...	...	...
2	Shrubs	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...	...
2'400						...					...

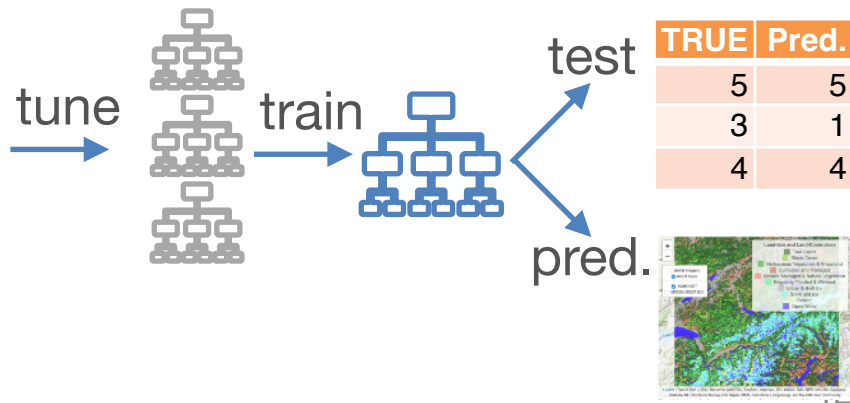
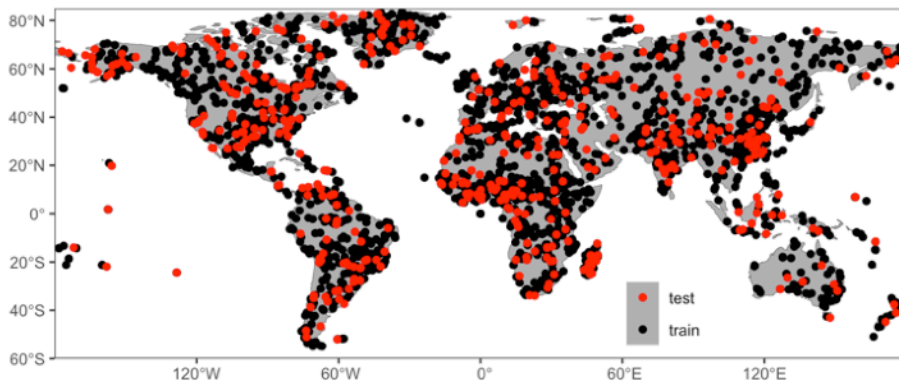
Reflectance bands 1 to 7 and Land surface temperature

Boosted Tree Classification of Land-Cover

Boosted classification trees: supervised classification (chapter 7.2)

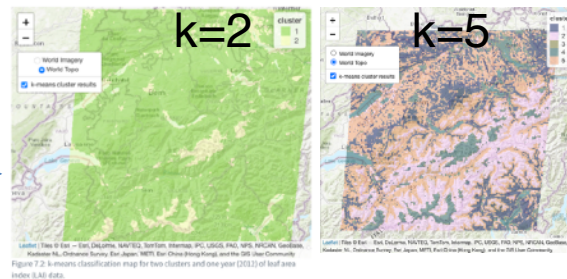
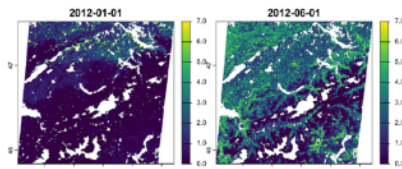
Pixel	Land Cover	B1 Feb	B1 Apr	B1 May	B1 June	...	B4 Feb	B4 Apr	B4 May	B4 June
1	Trees	...	...	...	...	...	...	...	...	...
2	Shrubs	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
2'400						...				

Reflectance bands 1 to 7 and Land surface temperature

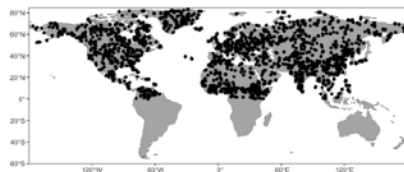


Chapter 7 in “Handful of Pixels” book

- 7.1 *Unsupervised* Land-Cover classification



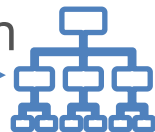
- 7.2 *Supervised* Land-Cover classification



tune



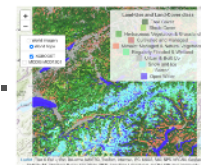
train



test

TRUE	Pred.
5	5
3	1
4	4

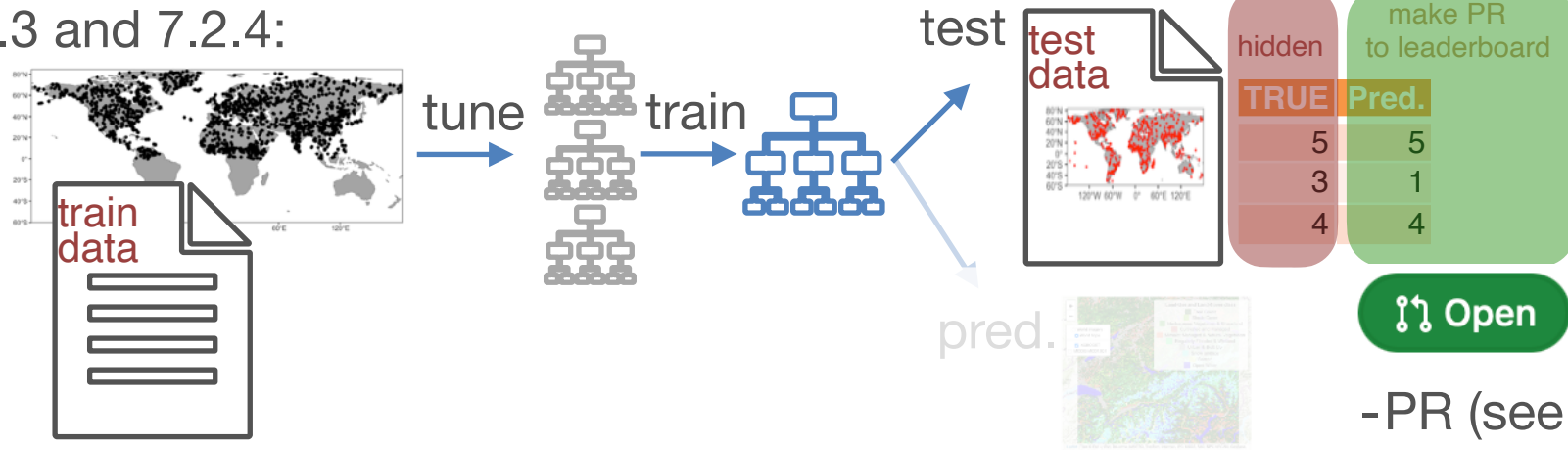
pred.



- Ex. 8.3: suggest improvements to 7.2
 - use your algorithm to predict 600 sites
- hand-in ==> Report + GitHub Pull Request

Tips for Exercise 8.3 in “Handful of Pixels” book

7.2.3 and 7.2.4:



Download data (see 7.2.3 “For Exercises”)



```
# train and test data for the competition was
# downloaded with `data-raw/download_competition_data.R`
# and prepared with `data-raw/compile_training_data.R`
# (see https://github.com/geco-bern/handfull_of_pixels/)
# and deposited on Zenodo.

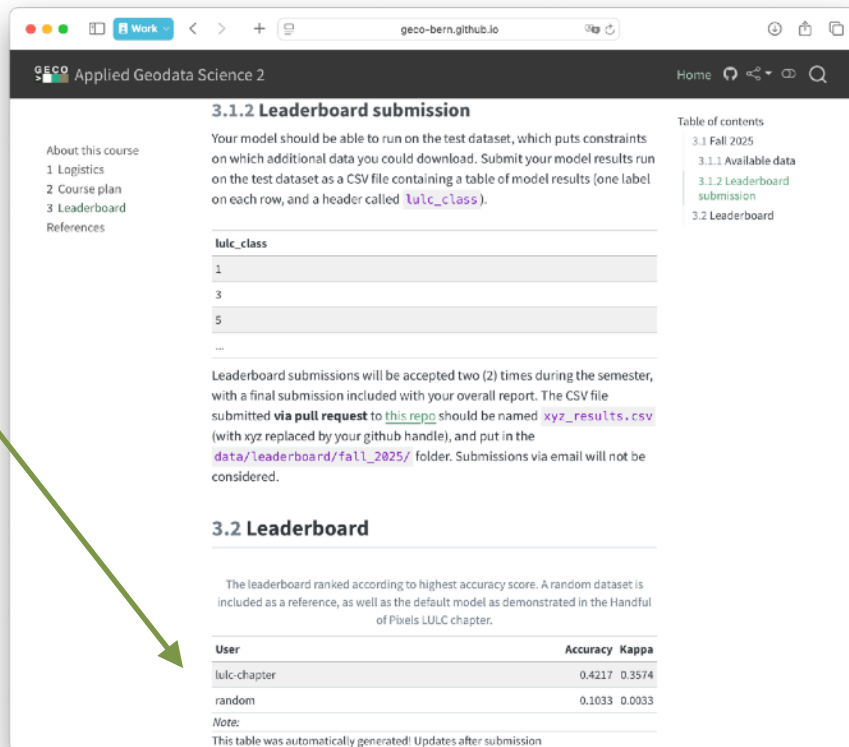
# see here how to directly load train and test data from Zenodo
download.file(
  url = "https://zenodo.org/records/8298491/files/test_data.rds?download=1",
  destfile = file.path(tempdir(), "test_data.rds")
)
download.file(
  url = "https://zenodo.org/records/8298491/files/training_data.rds?download=1",
  destfile = file.path(tempdir(), "training_data.rds")
)
```

- PR (see 7.2.4 “For Exercises”)
- .Rmd + .html

Share your results as Pull Request PR

Open

hidden		make PR to leaderboard	
TRUE	Pred.		
5	5		
3	1		
4	4		



The screenshot shows a web browser displaying the 'Applied Geodata Science 2' course page. The page includes a sidebar with navigation links, a main content area with instructions for submitting a pull request, and a table of contents. A green arrow points from the 'make PR to leaderboard' box in the top left to the '3.2 Leaderboard' section.

3.1.2 Leaderboard submission

Your model should be able to run on the test dataset, which puts constraints on which additional data you could download. Submit your model results run on the test dataset as a CSV file containing a table of model results (one label on each row, and a header called `lulc_class`).

lulc_class
1
3
5
...

Leaderboard submissions will be accepted two (2) times during the semester, with a final submission included with your overall report. The CSV file submitted **via pull request** to [this repo](#) should be named `xyz_results.csv` (with xyz replaced by your github handle), and put in the `data/leaderboard/fall_2025/` folder. Submissions via email will not be considered.

3.2 Leaderboard

The leaderboard ranked according to highest accuracy score. A random dataset is included as a reference, as well as the default model as demonstrated in the Handful of Pixels LULC chapter.

User	Accuracy	Kappa
lulc-chapter	0.4217	0.3574
random	0.1033	0.0033

Note:
This table was automatically generated! Updates after submission

Share your results as Pull Request PR

 Open

hidden

make PR
to leaderboard

TRUE	Pred.
5	5
3	1
4	4

Simple confusion matrices can be calculated using R using the `table` function comparing the true reference labels with the predicted values for our model.

For Book For Exercises

```
# NOTE: for the competition column 'LC1' was removed from
# the test data. Thus you can't compute the scores
# yourself.
#
# Instead upload your predictions from your best
# model as a pull request to
# https://github.com/geco-bern/agds2_course.
#
# To do so:
# a) save your results as *.csv:
# readr::write_csv(
#   x = dplyr::rename(test_results,
#                     lulc_class = .pred_class),
#   file = "~/Desktop/username_results.csv")
#
# a) fork the repository 'agds2_course',
# b) add and commit your results as
# 'data/leaderboard/fall_2025/username_results.csv',
# c) and open a pull request (PR) from your forked
# repository to the main repository at
# 'geco-bern/agds2_course'.
#
# Your predictions will then be added to the the leader
# board on the course website at
# https://geco-bern.github.io/agds2_course/leaderboard_fall_2025.ht
```

https://geco-bern.github.io/handfull_of_pixels/land_cover_classification.html#confusion-matrix-and-metrics

https://geco-bern.github.io/agds2_course/leaderboard_fall_2025.html

Earth observation sensors on satellites

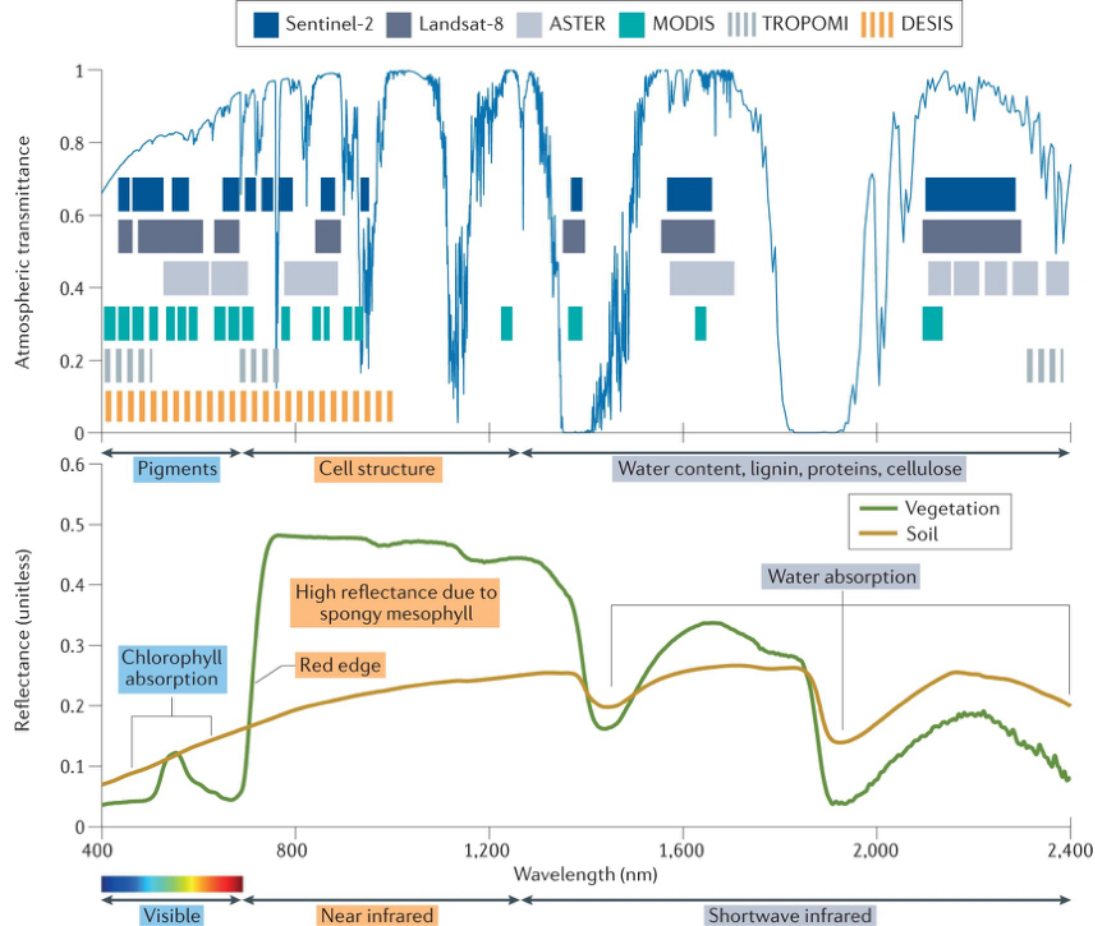


Figure 5.2: Figure from Zeng et al. (2022): 'Satellite sensors, the vegetation and the soil spectrum across wavelengths. Top: the spectral response range in the atmospheric window of a few widely used satellites¹⁰⁹. The coloured blocks and vertical lines illustrate the spectral band range or bandpass for each satellite sensor. Bottom: reflectance of vegetation and soil. These spectral features of vegetation and soil are the foundation of the rationale of vegetation indices and support the design of various vegetation indices.'